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Part 1

Issue/Need	Maintenance Type (Corrective / Adaptive / Perfective / Preventive)	Description	Impact
Login failure during peak hours	Corrective	Fixes system errors that prevent users from logging in when traffic is high	Students cannot access the system, causing delays and frustration
Queue display freezes	Corrective	Resolves bugs causing the display screen to stop updating	Students miss their turn, leading to confusion and inefficiency
Not mobile-friendly	Adaptive	Adjusts the system to work on mobile devices and different screen sizes	Limits accessibility for students using phones
Request for SMS notification	Perfective	Adds new features based on user requests	Improves user experience and convenience
Slow system performance	Perfective	Enhances system efficiency and speed	Long wait times and reduced productivity
Poor documentation	Preventive	Improves documentation to make maintenance easier	Difficult to update, increases risk of errors
Security concerns	Preventive	Fixes vulnerabilities before they are exploited	Risk of data breaches and system compromise
Browser compatibility issues	Adaptive	Updates system to work with new browser versions	Users may not access the system properly

Part 2

Task	Type	Priority	Proposed Solution	Tools/Technology
Fix login failure during peak hours	Corrective	High	Optimize server capacity, implement load balancing	Cloud hosting (AWS, Azure), Load balancer
Fix queue display freezing	Corrective	High	Debug and fix frontend/backend synchronization issues	JavaScript frameworks, WebSockets
Improve mobile compatibility	Adaptive	High	Use responsive design or mobile-first UI	Bootstrap, CSS Media Queries
Add SMS notification feature	Perfective	Medium	Integrate SMS API for alerts	Twilio, Nexmo
Improve system performance	Perfective	High	Optimize database queries and caching	MySQL optimization, Redis cache
Improve documentation	Preventive	Medium	Create clear technical and user documentation	Markdown, Confluence
Address	Preventive	High	Conduct	OWASP tools,

security vulnerabilities			security audit and apply fixes	SSL, Firewalls
Update browser compatibility	Adaptive	Medium	Update code to follow modern web standards	HTML5, updated JS frameworks

Part 3

Proposed Feature	Description	Reason	Expected Benefit
Mobile App with Real-Time Queue Tracking	A dedicated mobile app where students can view queue status, receive notifications, and manage appointments	Many students rely on mobile devices and need real-time updates	Faster service, improved accessibility, reduced crowding in offices